# Perioperative Management of Direct Oral Anticoagulants (DOAC's)

**There is a lack of evidence for the safety of surgery on these agents**

**Stop treatment before surgery**

- DOAC's have predictable pharmacokinetics and can usually be stopped close to the time of surgery
- Bridging therapy is not indicated

**Pre-op assessment team to confirm and communicate plan for timing of last dose of DOAC**

1. Check renal function
2. Use results (eGFR) to confirm plan for timing of last dose (see table below)

| Renal function (eGFR in ml/min) | Estimated half-life (h) | Timing of last dose of DOAC before surgery |
|---|---|---|
| Dabigatran | | |
| ≥80 | 13 | Two days |
| ≥50 to <80 | 15 | Three days |
| ≥30 to <50 | 18 | Four days |
| <30 | | Consult Haematologist |
| Rivaroxaban | 9 | Two days |
| Apixaban | 8 | Two days |
| Edoxaban | | |
| ≥30 | 10-14 | Two days |
| <30 | | Three days |

**Post-op:**

1. If adequate haemostasis achieved, give prophylactic LMWH at 6 – 8 hrs post-surgery if indicated by Trust VTE guideline
2. Surgeon to consider re-starting full anticoagulation on first post-operative day if haemostasis adequate and no contraindication (e.g. epidural in-situ).
3. **Following high risk procedures and in patients with an increased bleeding risk or in any situation where increased risk of bleeding is unacceptable, DOACs should not be re-introduced at full dose until at least 48 h post-procedure**